

ROS1 fusion-hotspot benchmark report

Study: luad_tcga_pan_can_atlas_2018

Abstract

This report analyzes ROS1 gene fusions from the luad_tcga_pan_can_atlas_2018 study, covering 10 fusion events. 8/10 fusions (80.0%) were in-frame. 8/10 fusions (80.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0222222$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for ROS1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 10 analyzed fusion events, identifying 5 distinct fusion partner genes. The most frequent partner was CD74, observed in 3 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 8/8 (100.0%) of in-frame fusions retained the domain, $p=0.0222222$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00599401.

Domain disruption

Disruption of the Fibronectin type III domain was tested with Fisher's exact test comparing in-frame fusions against all others; 7/8 (87.5%) of in-frame fusions disrupted the domain, $p=1$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.807419.

Cutpoint detection

Cutpoint detection scanned 10 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 866 aa (exon 17/18 boundary; permutation-corrected $p=0.031968$, statistically significant at $\alpha=0.05$). This is 586 aa from the nearest configured domain boundary at 280 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=8$) and "not_retained" ($n=2$). For Frame_status, retained: 8/8 (100.0%, 95% CI 67.6%-100.0%); not_retained: 0/0 (unavailable, 95% CI unavailable-unavailable). Welch's t-test comparing Tumor_variant_count between groups gave means of unavailable vs unavailable, $p=unavailable$.

Composite evidence score

Composite evidence score produced results for this run (5 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association produced results for this run; see the accompanying data for details.

Genomic position recurrence

Genomic position recurrence produced results for this run (6 row(s) in genomic_bin_recurrence, 1 row(s) in exact_position_recurrence, 1 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: ROS1 has no gene_pair configured; joint-partner analysis was skipped.

Mutation cooccurrence

Mutation cooccurrence reported: ROS1 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (10 row(s) in window_scan, 3 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CD74	3	0.3	0.2222282827095675	0.009290089627372218	0.4591039868475134	0.6755924351161198	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2838399301416053	1
FBXO9	1	0.1	0.2222282827095675	0.009290089627372218	0.44677353062063296	0.6755924351161198	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.22199036170757325	2
SLC34A2	2	0.2	0.2222282827095675	0.009290089627372218	0.230990546650226	0.6755924351161198	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.21962291411201218	3
EZR	2	0.2	0.2222282827095675	0.009290089627372218	0.18865598027127006	0.6755924351161198	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.21327272915516882	4
CLTC	2	0.2	0.2222282827095675	0.009290089627372218	0.0	0.6755924351161198	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18497433211447828	5

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
193	0	8	1	1	0.0	0.19999999999999998	0.6989700043360189
866	0	8	2	0	0.0	0.02222222222222222	1.6532125137753437
1750	1	7	2	0	0.0	0.06666666666666667	1.1760912590556813
1853	6	2	2	0	0.0	1.0	-0.0
1881	7	1	2	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

7	2
1	0

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
fibronectin	Fibronectin type III domain	10	1	0	9	0.0

domain_retention tables

frame_domain_contingency_table

8	0
0	2

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	10	8	0	2	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
36	8	10	0.8

frequency tables

Partner_gene_counts

Partner_gene	Event_count
CD74	3
CLTC	2
EZR	2
FBXO9	1
SLC34A2	2

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
6	117324000	117325000	5	EVT-TCGA-05-4426-01-9, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, EVT-TCGA-NJ-A7XG-01-5
6	117288000	117289000	1	EVT-TCGA-44-2665-01-7
6	117321000	117322000	1	EVT-TCGA-44-2665-01-3
6	117329000	117330000	1	EVT-TCGA-62-A46Y-01-10
6	117379000	117380000	1	EVT-TCGA-86-8278-01-6
6	117403000	117404000	1	EVT-TCGA-64-1680-01-8

exact_position_recurrence

chromosome	position	n_events	event_ids
6	117324415	5	EVT-TCGA-05-4426-01-9, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, EVT-TCGA-NJ-A7XG-01-5

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
1853	5	1	0	False

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1653	1853	200	6	4	6	2	None	0.13333333333333333	0.8750612633917001	EVT-TCGA-05-4426-01-9, EVT-TCGA-55-6986-01-4, EVT-TCGA-62-A46Y-01-10, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, ... (6 total)
1753	1853	100	5	5	5	3	None	0.4444444444444445	0.3521825181113625	EVT-TCGA-05-4426-01-9, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, EVT-TCGA-NJ-A7XG-01-5

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1781	1881	100	6	4	6	2	None	0.1333333333333333	0.8750612633917001	EVT-TCGA-05-4426-01-9, EVT-TCGA-44-2665-01-3, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, ... (6 total)
1803	1853	50	5	5	5	3	None	0.4444444444444445	0.3521825181113625	EVT-TCGA-05-4426-01-9, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, EVT-TCGA-NJ-A7XG-01-5
1828	1853	25	5	5	5	3	None	0.4444444444444445	0.3521825181113625	EVT-TCGA-05-4426-01-9, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, EVT-TCGA-NJ-A7XG-01-5
1831	1881	50	6	4	6	2	None	0.1333333333333333	0.8750612633917001	EVT-TCGA-05-4426-01-9, EVT-TCGA-44-2665-01-3, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, ... (6 total)
1853	1878	25	5	5	5	3	None	0.4444444444444445	0.3521825181113625	EVT-TCGA-05-4426-01-9, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, EVT-TCGA-NJ-A7XG-01-5
1853	1903	50	6	4	6	2	None	0.1333333333333333	0.8750612633917001	EVT-TCGA-05-4426-01-9, EVT-TCGA-44-2665-01-3, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, ... (6 total)
1853	1953	100	6	4	6	2	None	0.1333333333333333	0.8750612633917001	EVT-TCGA-05-4426-01-9, EVT-TCGA-44-2665-01-3, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, ... (6 total)
1853	2053	200	6	4	6	2	None	0.1333333333333333	0.8750612633917001	EVT-TCGA-05-4426-01-9, EVT-TCGA-44-2665-01-3, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, ... (6 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1831	1881	50	6	4	6	2	None	0.1333333333333333	0.8750612633917001	EVT-TCGA-05-4426-01-9, EVT-TCGA-44-2665-01-3, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, ... (6 total)
1653	1853	200	6	4	6	2	None	0.1333333333333333	0.8750612633917001	EVT-TCGA-05-4426-01-9, EVT-TCGA-55-6986-01-4, EVT-TCGA-62-A46Y-01-10, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, ... (6 total)
1828	1853	25	5	5	5	3	None	0.4444444444444445	0.3521825181113625	EVT-TCGA-05-4426-01-9, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, EVT-TCGA-NJ-A7XG-01-5

Per-event results (results.tsv)

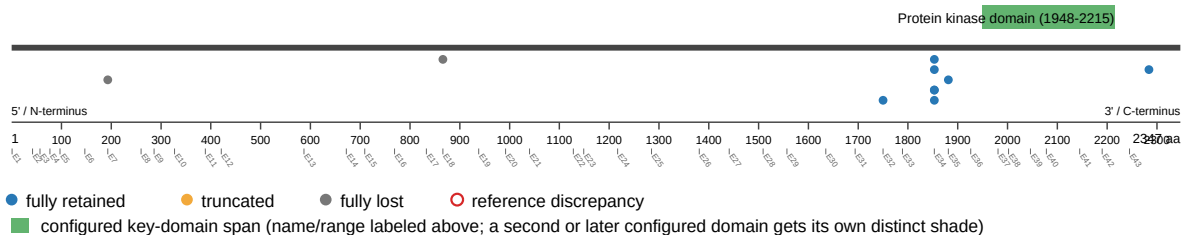
event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-64-1680-01-1	TCGA-64-1680-01		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117324415	34	1853	False	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NA CD74-ROS1 fusion	in-frame
EVT-TCGA-86-8278-01-2	TCGA-86-8278-01		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117324415	34	1853	False	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NA CD74-ROS1 fusion	in-frame
EVT-TCGA-44-2665-01-3	TCGA-44-2665-01		Non-Small Cell Lung Cancer	LUAD	CLTC-ROS1	CLTC	in-frame	three_prime	117321394	35	1881	False	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NA CLTC-ROS1 fusion	in-frame
EVT-TCGA-55-6986-01-4	TCGA-55-6986-01		Non-Small Cell Lung Cancer	LUAD	EZR-ROS1	EZR	in-frame	three_prime	117324415	34	1853	False	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NA EZR-ROS1 fusion	in-frame
EVT-TCGA-NJ-A7XG-01-5	TCGA-NJ-A7XG-01		Non-Small Cell Lung Cancer	LUAD	EZR-ROS1	EZR	in-frame	three_prime	117324415	34	1853	False	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NA EZR-ROS1 fusion	in-frame
EVT-TCGA-86-8278-01-6	TCGA-86-8278-01		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	unknown	five_prime	117379059	17	866	False	865	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain; Fibronectin type III domain		NA ROS1-CD74 fusion	frameshift

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-44-2665-01-7	TCGA-44-2665-01		Non-Small Cell Lung Cancer	LUAD	ROS1-CLTC	CLTC	in-frame	five_prime	1172886	43	2284	False	2283	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain; Fibronectin type III domain			NA ROS1-CLTC fusion	in-frame
EVT-TCGA-64-1680-01-8	TCGA-64-1680-01		Non-Small Cell Lung Cancer	LUAD	ROS1-FBXO9	FBXO9	unknown	five_prime	1174031	6	193	False	192	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain; Fibronectin type III domain		NA ROS1-FBXO9 fusion	frameshift
EVT-TCGA-05-426-01-9	TCGA-05-426-01		Non-Small Cell Lung Cancer	LUAD	SLC34A2-ROS1	SLC34A2	in-frame	three_prime	1173244	34	1853	False	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NA SLC34A2-ROS1 fusion	in-frame

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-62-A46Y-01-10	TCGA-62-A46Y-01		Non-Small Cell Lung Cancer	LUAD	SLC34A2-ROS1	SLC34A2	in-frame	three_prime	117329446	32	1750	False	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NA SLC34A2-ROS1 fusion	in-frame

domain retention outliers

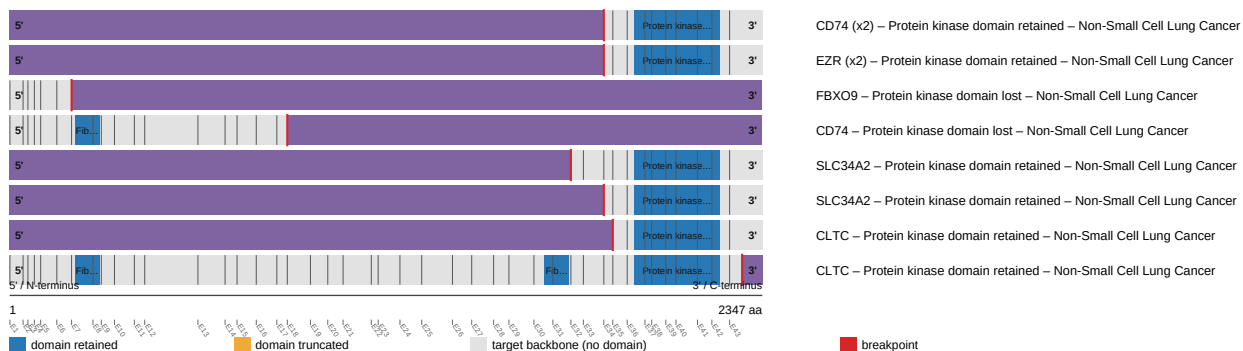
ROS1 domain-retention track



fusion schematic

ROS1 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (ROS1 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.